CONSOLIDATED GROWTH PROGRAM

Sun Gold & Black Cherry Tomatoes

Taste › Quality › Volume

Yara Fertiliser Inventory: Calcinit · Krista MKP · Krista SOP · Krista MgS · Rexolin APN
Substrate: 50/50 Peat Moss + Pine Fiber
Seeding: 16 March | First Transplant: 15 April | Greenhouse: 15 May

Synthesised from four independent AI research programs — April 2026
Assumption: low-to-moderate alkalinity water. Treat as v1 until water analysis is complete.

1. Guiding Principles

This program synthesises four independent research programs into a single actionable plan, choosing the best-supported recommendation at each decision point. Every parameter is tuned for your priority order: taste first, then quality, then volume.

1.1 Flavor Levers

- **Staged K:N ratio** — from ~1:1 early to ~1.9:1 at harvest. Potassium drives sugar accumulation, acid balance, and volatile compound synthesis.
- **Moderate EC elevation during fruiting** — capped at 2.8–3.1 dS/m (not higher, because your only Ca source is Calcinit, which ties calcium to nitrogen). Cherry tomatoes respond to EC with higher Brix, but your fertiliser inventory sets a practical ceiling.
- **Controlled evening dryback** — let substrate tension rise 1–1.5 kPa above the irrigation trigger by end of day. Research shows moving from −30 to −60 hPa improved TSS 8–12% in cocktail tomatoes, but also cut yield ~10%. Moderate dryback gets most of the flavour benefit with less yield loss.
- **Day/night temperature differential of 4–7°C** — cooler nights promote sugar transport from leaves to fruit and support volatile synthesis.
- **Drain-EC monitoring** — research found fruit Brix strongly correlated with cumulative drain EC from anthesis to harvest. Drain EC is more informative than feed EC alone.

1.2 Key Constraints from Your Inventory

- **Ca is tied to N:** Calcinit is your only calcium source. You cannot raise Ca independently of N. This limits maximum EC and means BER management relies heavily on even irrigation, not just more calcium.
- **K is tied to sulfate:** Krista SOP is your main K driver. Very high K recipes would overload sulfate. Recipes are capped to keep S reasonable.
- **No potassium nitrate or calcium chloride:** Classic high-K greenhouse tomato recipes use KNO_3 and $CaCl_2$. Without them, the program compensates with mild dryback for flavour steering.

1.3 Tank Setup

Tank A: Calcinit only.

Tank B: Krista MKP + Krista SOP + Krista MgS + Rexolin APN.

> *Never combine calcium nitrate with phosphates or sulfates in concentrate — causes calcium sulfate/phosphate precipitation that clogs drippers. Yara explicitly places APN in Tank B. Keep APN stock out of light and use within 1–2 weeks.*

2. Climate, Light & Irrigation Program

Stage	Day °C	Night °C	PPFD / Hours	DLI target	Feed pH	Target EC
1. Plugs (Now–14 Apr)	21–22	18–19	200–220 / 18h	~13 mol	5.5–5.8	1.8–2.0
2. 13cm settle (15–21 Apr)	24 (48h) then 21	18	230–250 / 18h	~15 mol	5.5–5.8	2.0–2.2
3. 13cm build (22 Apr–8 May)	20–21	17–18	250–300 / 18h	~16–19	5.5–5.8	2.2–2.4
4. Harden (9–15 May)	18–19	16	220 / 16h	~12–13	5.5–5.8	85–90% of stage 3
5. GH establish (15 May–flowers)	22–24	17–18	Natural sun	≥20 mol	5.4–5.8	2.3–2.5
6. Flower–first colour	22–25	17–18.5	Natural sun	≥25 mol	5.4–5.7	2.6–2.8
7. Main harvest	22–26	17–19	Natural sun	≥25 mol	5.4–5.7	2.8–3.1 bright 2.6–2.8 dull

Hardening week: *Drop temperature, shorten photoperiod, and reduce feed to 85–90% of stage 3 levels for 5–7 days before greenhouse move. This prepares plants for the stress of transplanting and outdoor conditions.*

2.1 Irrigation & Tensiometer Targets

Stage	Trigger kPa	Evening dryback	Drain target	Volume guide
1. Plugs	—	Slight surface dryback	—	Keep evenly moist, never waterlogged
2. 13cm settle	1.5–2.0	2.5–3.0 kPa	—	100–150 mL/pot/day rising to 150–300 mL
3. 13cm build	2.0–2.5	3.0–4.0 kPa	10–20%	250–500 mL/pot/day in 2–4 pulses
5. GH establish	2.0–3.0	3.5–4.5 kPa	10–20%	0.6–1.0 L/plant/day in 2–4 pulses, rising to 1.0–1.8 L
6. Flower–colour	3.0–3.5	4.0–5.0 kPa	10–20%	0.8–1.6 L/plant/day in 4–8 pulses
7. Harvest	3.0–3.5	4.0–5.0 kPa overnight	10–20%	1.5–3.0 L/plant/day in 6–12 pulses. Last irrigation 2–3h before sunset

Variety-specific: *Keep Sun Gold on the wetter side of these ranges (lower kPa trigger), especially from colour break onward — it is very prone to splitting. Black Cherry can handle the drier end for extra flavour concentration.*

3. Fertigation Recipes

All amounts are grams per 1,000 L of final irrigation water. At a 1:100 injector, divide by 10 for g/L stock.

Stage	Calcinit	MKP	SOP	MgS	APN	EC target	Approx ppm N/P/K/Ca/Mg
1. Plugs	650	110	200	330	25	1.8–2.0	100 / 25 / 115 / 123 / 32
2. 13cm settle	805	120	300	415	25	2.0–2.2	125 / 27 / 159 / 152 / 40
3. 13cm build	935	115	400	465	25	2.2–2.4	145 / 26 / 205 / 177 / 45
4. Harden	~800	~100	~340	~400	25	~2.0	85–90% of stage 3
5. GH establish	1000	115	450	470	25	2.3–2.5	155 / 26 / 230 / 189 / 45
6. Flower–colour	1000	110	530	500	25	2.6–2.8	155 / 25 / 260 / 189 / 48
7. Harvest	970	105	600	520	25	2.8–3.1	150 / 24 / 286 / 183 / 50

3.1 What These Recipes Are Doing

The progression follows the classic tomato steering principle: vegetative balance early, generative balance later.

- **N drops from 155 to 150 ppm at harvest** — avoids excessive vegetative growth that dilutes flavour and increases disease pressure.
- **K climbs from 115 to 286 ppm** — the strongest lever for sugar, acid, and volatile compound accumulation. K:N moves from ~1.15:1 to ~1.9:1.
- **Ca stays high (183–189 ppm) throughout fruiting** — BER prevention and firm fruit texture. Since Ca is tied to N via Calcinit, this is near the practical maximum.
- **P decreases slightly at harvest (24 ppm)** — phosphorus demand does not increase during fruiting. Reducing MKP slightly avoids pushing P beyond what the plant needs.
- **Mg maintained at 45–50 ppm** — high K can antagonise Mg uptake. Keeping Mg steady prevents interveinal yellowing of older leaves.

3.2 Rexolin APN Micronutrient Contribution

At 25 g/1,000 L, Rexolin APN delivers approximately:

Fe	Mn	Zn	B	Cu	Mo	Notes
~1.5 ppm	~0.6 ppm	~0.33 ppm	~0.23 ppm	~0.06 ppm	~0.06 ppm	Nordic APN (0.9% B version). If Norwegian 1.1% B, B is ~0.28 ppm

APN dose rationale: *Yara's generic vegetable rate is 14 g/1000 L. Cornell's tomato-specific targets run higher (~2 ppm Fe). 25 g is a middle ground: more than the generic vegetable rate, slightly below the 30 g one agent used, and puts Fe at ~1.5 ppm which is reasonable for peat/pine fiber where Fe availability can drop if root-zone pH drifts upward.*

4. Critical Management Rules

4.1 Pine Fiber Nitrogen Immobilisation

Wood-fiber substrates can temporarily lock up plant-available nitrogen, and this effect gets more pronounced above ~30% wood content. Your 50/50 mix is well above that threshold. Watch for:

- Pale or yellowish new growth 7–14 days after each transplant
- **If this appears:** temporarily increase Calcinit by 10–15% for about one week, rather than pushing MKP or SOP
- **Pre-wet every new pot with nutrient solution, not plain water** — this gives the substrate an initial nutrient charge

4.2 Root-Zone pH Management

Your peat/pine fiber mix starts very acidic (pH 3.5–4.5). The fertigation will buffer this upward, but pH can drift more easily in wood-containing substrates than in peat-only mixes.

- Check pour-through pH and EC weekly after each transplant
- Target root-zone pH: 5.6–6.2
- If root-zone pH rises above 6.2: push feed pH to the low end (5.4–5.5)
- If root-zone pH drops below 5.5: raise feed pH toward 6.0–6.2 briefly, or surface-apply a small amount of dolomitic lime
- APN uses DTPA-chelated iron, which remains available up to about pH 6.5 — one reason it suits soilless systems where pH may drift

4.3 Calcium & BER Prevention

Blossom End Rot is the #1 quality risk for your crop. The root cause is almost always inconsistent water supply, not calcium deficiency per se. Calcium is immobile in the plant and travels only via the transpiration stream.

- **Primary defence:** steady, even irrigation via tensiometer. Avoid big wet/dry swings.
- **Secondary defence:** keep Calcinit high enough (183–189 ppm Ca at harvest). Your recipes already maximise this within the N constraint.
- **If BER appears despite even irrigation:** the nutrient-side fix is a second Ca source (e.g. calcium chloride). Your current inventory doesn't include one, so plan for it as a contingency purchase.
- **Avoid high ammonium nitrogen** — it competes with Ca uptake. Calcinit provides nitrate-N, which is correct.

4.4 Drain-EC Monitoring

Drain EC is your most important ongoing measurement for flavour management:

- Measure drain EC and pH weekly from every set of pots
- **If drain EC exceeds feed EC by more than 1.0–1.5 dS/m:** increase irrigation frequency to flush salts. At EC 2.8–3.1 feed, salts accumulate quickly in peat.
- Track cumulative drain EC from anthesis to harvest — this correlates with Brix better than feed EC alone
- If a full flush is needed: irrigate with plain pH-adjusted water (pH 5.5) at 2× normal volume for one cycle

4.5 Pollination

- Bumblebees (Bombus terrestris) are the gold standard for greenhouse tomato pollination
- **If no bumblebees:** manually vibrate flower clusters every other day around midday using an electric toothbrush or vibrating wand
- Optimal conditions: ~70% RH, 18–29°C at blossom height, below 30°C
- Above 80% RH pollen sticks together; below 60% RH stigma dries out

4.6 Pruning & Training

Train both varieties to a single leader stem. Remove all suckers when they are less than 5 cm. Remove lower leaves below the lowest ripening truss for air circulation. Sun Gold is exceptionally vigorous — be diligent with sucker removal. Consider topping plants at 6–8 trusses to focus energy on fruit quality.

5. Harvest & Quality Monitoring

5.1 Harvest Timing for Peak Flavour

Sun Gold: Wait for the final colour shift from golden to pale apricot-orange. This last-minute change signals peak sugar and volatile development. Flavour develops early in this variety, but the final transformation is what makes it exceptional. Sun Gold is very prone to splitting when fully ripe and well-watered — the controlled dryback strategy helps, but harvest promptly at peak colour.

Black Cherry: Harvest when fruit is deep mahogany-purple with slight give when gently squeezed. The characteristic smoky sweetness only develops at full ripeness.

Never refrigerate. Store at room temperature (above 13°C). Temperatures below ~12°C destroy the enzyme responsible for tomato flavour volatiles.

5.2 Brix Targets

Measure with a handheld refractometer at consistent time of day, from fully ripe fruit:

Sun Gold target: 8–10+ °Brix (exceptional specimens can reach 11–12)

Black Cherry target: 7–9+ °Brix (these tend to run slightly lower than Sun Gold)

If Brix is below target: check that drain EC has been maintained, K levels are at the harvest recipe, and dryback is being applied. Increase SOP slightly if room within EC band.

6. Injection System Setup

220 plants × 4 L/hr drippers = 880 L/hr peak flow (14.7 L/min / 3.9 GPM). This fits comfortably within the Newtry injector range of 20–2,500 L/hr.

6.1 Equipment List

Injector 1 — Tank A (Calcinit): Newtry 0.4–4% model (1:250 to 1:25), set to 1% (1:100). This is the standard greenhouse fertigation ratio.

Injector 2 — Tank B (MKP + SOP + MgS + APN): Same model, Newtry 0.4–4%, also set to 1% (1:100).

Injector 3 — Acid: Newtry 0.2–2% model (1:500 to 1:50). This gives finer control at low injection rates. Run at approximately 0.3–0.5% — exact setting determined by titration of your water.

Tank A: 200 L barrel (Calcinit stock solution).

Tank B: 200 L barrel (MKP + SOP + MgS + APN stock solution).

Acid tank: 25 L can (diluted phosphoric acid). Consider upgrading to 50 L for less frequent refills.

Filter: 100–120 mesh inline filter before the first injector.

Spare parts: Buy one spare Newtry injector body + seal kit upfront. User reviews report occasional seal failures after 6–12 months under continuous use.

6.2 Plumbing Layout

Install the three injectors in series on the main water line. Water flows through each in sequence:

Water supply → Filter (100–120 mesh) → Injector 1 (Tank A) → Injector 2 (Tank B) → Injector 3 (Acid) → Main line → Drippers

- Install a bypass valve around all three injectors so you can flush with plain water when needed
- Each injector mounts vertically — pump body on top, suction tube dropping down into the barrel/can
- Install a ball valve on each barrel's suction line to isolate a tank without disconnecting
- The arrow on the Newtry pump body indicates flow direction — match it to your pipe flow
- 3/4” NPT connections. You may need 3/4” NPT-to-hose-thread adapters depending on your pipe

Start-up procedure: *After installation, open the water inlet valve slightly, then press the exhaust valve on each injector to purge air. Release when a continuous flow of water overflows the exhaust valve. The injector will not dose correctly with air in the system.*

7. Stock Tank Concentrations (1:100 Injection)

At 1% (1:100) injection ratio: multiply the fertigation recipe (g/1000 L final water) by 0.1 to get g/L of stock solution. The tables below give both stock concentration and total kg per 200 L barrel fill.

7.1 Tank A — Calcinit Only

Dissolve in warm water (25–30°C), stir until fully clear. Calcinit solubility is ~1,200 g/L — your concentrations are well within range.

Stage	Calcinit g/L stock	Total kg per 200L	Barrel longevity*
1. Plugs	65.0	13.0 kg	N/A (hand watering)
2. 13cm settle	80.5	16.1 kg	N/A (hand watering)
3. 13cm build	93.5	18.7 kg	N/A (hand watering)
5. GH establish	100.0	20.0 kg	~55 days
6. Flower–colour	100.0	20.0 kg	~42 days
7. Harvest	97.0	19.4 kg	~30–36 days

** Barrel longevity based on: GH establish ~1 L/plant/day (220 L/day), Flower ~1.5 L/plant/day (330 L/day), Harvest ~2.5 L/plant/day (550 L/day). At 1% injection the barrel consumes 1% of total water flow.*

7.2 Tank B — MKP + SOP + MgS + APN

Dissolve in order: MgS first (most soluble), then MKP, then SOP (least soluble — use warm water), then APN last. Stir thoroughly. Keep barrel out of direct sunlight (APN is light-sensitive). Stir every few days. Use within 1–2 weeks.

Stage	MKP g/L	SOP g/L	MgS g/L	APN g/L	Total g/L	Total kg per 200L
1. Plugs	11.0	20.0	33.0	2.5	66.5	13.3 kg
2. 13cm settle	12.0	30.0	41.5	2.5	86.0	17.2 kg
3. 13cm build	11.5	40.0	46.5	2.5	100.5	20.1 kg
5. GH establish	11.5	45.0	47.0	2.5	106.0	21.2 kg
6. Flower–colour	11.0	53.0	50.0	2.5	116.5	23.3 kg
7. Harvest	10.5	60.0	52.0	2.5	125.0	25.0 kg

SOP solubility note: *Krista SOP dissolves at ~120 g/L at 20°C. Your maximum stock concentration is 60 g/L SOP (harvest stage), well within limits. Use warm water when mixing to speed dissolution.*

7.3 Acid Tank — Phosphoric Acid (25 L Can)

Use food-grade 85% phosphoric acid (H_3PO_4). This is the safest common acid for small greenhouse operations. It adds a small amount of P, which is acceptable since MKP in Tank B is already conservative.

Preparing the acid stock

SAFETY: Always add acid to water, never water to acid. Wear gloves and splash goggles.

1. Fill the 25 L can with approximately 22 L of clean water
2. Slowly pour in approximately 3 L of 85% phosphoric acid while stirring
3. Top up to 25 L with water — this gives a ~10% H_3PO_4 working stock
4. Label the can clearly: “10% PHOSPHORIC ACID — CORROSIVE”

Calibrating the injection rate

Your water is pH 7.8, but the alkalinity (buffering capacity) determines how much acid you actually need. You must titrate-test your specific water:

5. Fill a bucket with exactly 10 L of your raw water. Measure pH (should read ~7.8)
6. Using a syringe, add your 10% acid stock drop by drop, stirring after each addition

7. Record how many mL it takes to reach pH 5.5
8. Multiply by 100 to get mL per 1,000 L — this is your acid dose
9. Set the Newtry 0.2–2% injector to deliver that amount at your flow rate

Typical starting point: *Most moderate-alkalinity water (2–3 meq/L HCO_3^-) needs about 0.3–0.6 mL of 85% H_3PO_4 per litre to drop from pH 7.8 to ~5.5. With the 10% stock, that translates to roughly 2.5–5 mL stock per litre of final water, which means the injector set at about 0.3–0.5%.*

Acid tank longevity

At 0.4% injection and 550 L/day total water (harvest stage): 550 × 0.004 = 2.2 L/day of acid stock consumed. The 25 L can lasts approximately 11 days. Consider upgrading to a 50 L container if frequent refills become inconvenient.

7.4 Operational Notes

- **Always measure final pH and EC at the dripper after setup** — not just barrel concentrations. Adjust injector percentages based on actual output measurements.
- **Flush the acid injector with clean water after each irrigation cycle** — acid stock (pH ~1.3) will wear seals faster than fertiliser. Inspect seals monthly.
- **Check the inline filter weekly** — undissolved fertiliser particles or substrate debris can clog it.
- **If any barrel runs dry while irrigating, the injector simply stops dosing** — plants receive plain water. Monitor barrel levels to avoid undiluted irrigation events.
- **Newtry injectors are water-powered (no electricity needed)** — they work on the water pressure in your line (0.3–6 bar / 4–87 PSI). Verify your greenhouse water pressure is within this range.
- **To change recipe between growth stages:** drain and rinse the barrel, then mix the new stock. Or simply let the old stock run low and top up with freshly mixed solution at the new concentration.

Reliability note: *Newtry injectors are a good value option, but user reviews report occasional seal failures at 6–12 months under continuous commercial use. For a 220-plant operation, have a spare unit on hand. For maximum reliability, Dosatron D14MZ2 or MixRite TF-10 are the professional standard (€300–500 each vs. ~€50–80 for Newtry).*

8. Weekly Monitoring Checklist

1. Measure feed solution EC and pH before every fertigation batch
2. Measure pour-through / drain EC and pH at least weekly
3. Measure final pH and EC at the dripper — verify injector output matches targets
4. Read tensiometer values; calibrate irrigation trigger and evening dryback
5. Record min/max temperature and humidity daily
6. Scout for pests (whitefly, aphids, thrips, spider mites) and disease (botrytis, septoria)
7. Remove suckers, prune lower leaves, train stems
8. Check Brix with refractometer once fruits begin ripening
9. Calibrate pH and EC meters monthly
10. Inspect tensiometers for air bubbles and refill water if needed
11. Check stock barrel levels — Tank A, Tank B, and acid can. Note consumption rate
12. Check inline filter for debris; clean if needed
13. Inspect injector seals monthly; listen for clicking sound confirming operation

9. Critical Next Steps

1. **Get a water analysis done:** alkalinity, EC, Ca, Mg, Na, Cl, bicarbonate, SO_4. This is the single most important upgrade. Your water chemistry will change the fertiliser recipe and the acid injection rate. All four source programs flagged this.
2. **Confirm your APN bag is the Nordic version:** check the B% on the label. If 0.9% (Swedish) or 1.1% (Norwegian), the 25 g/1000L dose is fine. If a different formulation, adjust accordingly.
3. **Run the acid titration test on your water:** follow section 7.3 to determine exact injection rate for the Newtry 0.2–2% acid injector.
4. **Verify greenhouse water pressure:** Newtry injectors require 0.3–6 bar (4–87 PSI). Measure at the greenhouse tap before ordering.
5. **Consider stocking calcium chloride as a contingency:** if BER appears despite even irrigation, it's the only way to add Ca without more N.
6. **Order spare Newtry injector + seal kit:** for a commercial 220-plant operation, a failure during peak season is costly. Have a backup ready.

This program assumes low-to-moderate alkalinity source water with little Ca/Mg. Treat it as v1 until you have a water analysis. All gram amounts are starting points — always measure actual solution EC and adjust. Your specific water quality, microclimate, and substrate batches will require fine-tuning. The fertigation rates are for final irrigation water at the dripper; for stock tanks, adjust to your injector ratio.